It checks a ship design and shows its proof

Built overnight on the team's app, Dynamis SCC. You give it the ship's files. It finds the rules that apply, checks each one, and answers pass, fail, conflict, or can't tell. Every answer shows where its numbers came from.

> The AI only reads and points. Plain code decides every pass and fail. The engineer has the final say. When it isn't sure, it asks instead of guessing.

How it works

1 Read
Drawings, specs, door and stair lists, IFC and 3D models. Each value keeps its file, page, spot and exact words.

→

2 Double-check
A number counts only when two different readings agree. One reading alone can't decide anything.

→

3 Pick the rules
From the ship's type, size, passengers, voyage and dates, it lists which rules apply, and why.

→

4 Check
Code tests each rule. Anything it can't prove becomes a question with the rule's clause.

→

5 Prove
Each run makes a proof file. Anyone can re-check every number and every quote.

Results on our test ship

SYN-1, a made-up cruise ship with 12 planted mistakes

40 of 41

expected results right. The one miss is a "can't tell": the evacuation time waits for the engineer to confirm two numbers.

0 false passes

It never said "pass" when the right answer was fail, conflict or can't tell.

12 of 12

planted mistakes caught: too-narrow doors and stairs, a wrong scale, files that disagree, missing values, a stale schedule.

0 wrong values

Of the 296 values in the files, 206 were read right and none wrong. The rest it asks for instead of guessing.

8 of 9

rule limits match the official IMO texts word for word. Two did before tonight.

28 of 28

doors right on the real RCRV 3D model, and 193 of 193 seeded faults caught. Unchanged by tonight's work.

New things the team can show

A drawing with the wrong scale

The title block says 1:100. Measured from its own frame marks, the sheet is 1:200. Nothing is measured off it until someone fixes it.

Revision B

Revision B replaces revision A's values and lists what changed: stair B, 850 → 1000 mm. The schedule still says 850, so that shows as a conflict.

The proof file

One zip per run. Change one byte and the check fails. Re-running the checks on it gives the same answers.

What if, and sister hulls

Change a date or a number and see which rules switch on, and which results move. Nothing is saved until you confirm.

Bridge view, on the drawing

The engineer clicks the eye, the obstruction and the bow on a measured profile. Code works out how much sea is hidden ahead of the bow, and draws it on the sheet.

The AI teaches the reader

When the AI finds a label the reader doesn't know, an engineer approves it once. From then on, code reads it on its own.

On the real RCRV drawing

a CAD plot with drawn-on text and no text layer, 12 pages; OCR, with second readings by the AI stand-in

1:196.7, not 1:96

The note says 1/8″ = 1′ -0″. Measured on the PDF from 10 deck heights, the sheet is half that size. Anything scaled off it with the note would be wrong by half, so the app flags it.

24.0 in, two ways

The frame marks and the deck heights give the same frame spacing on their own. That's how the app knows its own measurement is right.

7 values confirmed

LOA 199′ -6″, beam 41′ -0″, draft 12′ -6″ and more. OCR read them, and an independent transcription agreed, so each counts as proven (tier B).

The product, in one line

Product **It finds the rules that apply to your ship, proves each answer on your own drawings, and turns everything it can't prove into a question with the clause attached.**

Numbers: docs/overnight/SCOREBOARD.md (tools/scoreboard.py) and tools/accuracy_report.py, both run with no AI calls. Test ship and its answer key: samples/meyer-style-v1.

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Run it, decide, and what's next

Everything is in local commits on the branch **overnight-prototype**. Nothing is pushed. No real AI calls were made. 25 AI questions were answered by closed helpers that saw only the question, the way the real AI would.

Run it

```
# set up once
python -m venv .venv
.venv/bin/pip install -r backend/requirements.txt
# start the app: http://127.0.0.1:8000
cd backend
../.venv/bin/python -m uvicorn app.main:app
# score it on the test ship (no AI needed)
.venv/bin/python tools/scoreboard.py
# make a proof file, and check it
curl localhost:8000/api/proof/card -o proof.zip
.venv/bin/python tools/verify_proof.py proof.zip
```

Turn on the real AI tomorrow

- Put the key in `.env` and set `AI_MODE=record`. Questions already answered replay for free. New ones go to the API and are saved.
- Offline mode stays the default, so nothing leaves the machine by accident.
- `docker compose up` is written but not tested. There's no Docker on this machine.

Where things are

- `docs/ARCHITECTURE.md`: how it fits together, and every endpoint
- `docs/overnight/`: the plan, the scoreboard, and the three reviews
- `docs/rules/imo/`: the official rule texts the limits are checked against

Decide before Sunday

we didn't change these on our own

Item	What it does now	Your call
Door depth on the old test profiles	The team's rule is kept: door openings are taken as not deep. It's now a named, approved assumption. Ship-card mode checks depth exactly.	Keep it, or drop it?
32 in or 815 mm	Inches decide, as the team chose: 813 mm passes, with a note. A reviewer suggests using the stricter figure.	Keep inches, or use 815 mm?
Liferafts "100% each side"	This rule can't be traced to a clause yet, because the cargo-ship text isn't in our rule texts. In ship-card mode it runs only for cargo-ship lifesaving.	Find the text, or limit it to the RCRV?
Persons on board	The old raft check adds crew and science together. The ship card never adds them up; it asks instead.	Ask, as the card does?
IMO text in the repo	Short excerpts of the adopted IMO texts sit in `docs/rules/imo/`, so the limits can be checked word for word. IMO bars commercial reuse of its texts.	Keep them local only before the repo goes public?
Evacuation time	The engineer states the total and E+L from the analysis. A table row can belong to a case that doesn't govern, so code doesn't read it.	OK for the demo?

Known limits

- The test ship is ours, made to look like a yard package. Meyer gave us no real passenger-ship files.
- Drawings with drawn-on text are read by OCR (18 of 26 values on our test sheet). Low-quality scans aren't read yet. The app says "can't tell" instead of guessing.
- Values beside door and stair tags are linked by position, so they count as one reading. They can raise a conflict, but can't pass anything alone.
- The rule texts are the adopted IMO versions. Later amendments are listed where we checked them.
- The bridge-view check is tested on test data only. Next, try it on the RCRV profile, page 6.
- The web app got small updates only. The new features are in the API.

Ask Meyer at the Q&A, 09:00

1. Which rule families matter most to you after fire safety and lifesaving?
2. What files would a pilot get: GA PDFs, NWD or IFC models, schedules exported from your design tool?
3. Would a proof file help in class approval with DNV or Lloyd's Register, and what should it show?
4. When two drawings disagree today, who decides, and where is it recorded?

Built 7 Nov 2026 from the team's repo, the reading-design and product-direction PDFs, and the team rules. Reviews: docs/overnight/review/ (QA, rules, product). Business notes: research_notes/Product direction and hackathon strategy/business_model.md.

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